

Comprehensive Sanity Checks

GBIF Gap Analysis - Data Quality & Readiness Assessment

February 13, 2026

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Overview

This notebook provides comprehensive quality checks for the GBIF occurrence data, validating readiness for **spatial**, **temporal**, and **taxonomic** gap analysis.

Run this after: Scripts 01-06 (ingestion + derived summaries)

Purpose:

1. Verify all required files exist and are properly formatted
2. Identify data quality issues before gap analysis
3. Characterize spatial, temporal, and taxonomic coverage
4. Flag potential biases that may affect gap interpretation

Last run: 2026-02-13 10:55:42.475988

1. File Inventory & Validation

1.1 Core Processed Files

Table 1: Core Processed Files

file	description	status	size_mb
grids_10km.gpkg	EEA 10km reference grid	OK	1.63
grids_50km.gpkg	EEA 50km reference grid	OK	6.00
taxa_reference_current.rds	National taxonomy backbone	OK	7.54
cube_manifest.csv	Cube file metadata	OK	0.00
cube_totals_by_basisOfRecord.csv	Occurrence totals summary	OK	0.00

1.2 Derived Summary Files (Script 06a)

Table 2: Core Derived Files (from 06a)

file	description	status	size_mb	rows
cell_summary_10km.csv	Occurrences per cell (10km)		1.11	27,816
cell_summary_50km.csv	Occurrences per cell (50km)		0.13	3,452
time_summary_10km.csv	Occurrences per month (10km)		0.39	9,991
time_summary_50km.csv	Occurrences per month (50km)		0.38	9,991
cell_time_summary_10km.csv	Cell $\times$ month matrix (10km)		174.84	3,985,642
cell_time_summary_50km.csv	Cell $\times$ month matrix (50km)		26.38	582,271
order_cell_summary_10km.csv	Order $\times$ cell (10km)		71.26	1,424,257
order_cell_summary_50km.csv	Order $\times$ cell (50km)		8.93	173,410
order_time_summary_10km.csv	Order $\times$ month (10km)		29.24	613,421
order_time_summary_50km.csv	Order $\times$ month (50km)		29.18	613,421
family_time_summary_10km.csv	Family $\times$ month (10km)		92.41	1,616,590
family_time_summary_50km.csv	Family $\times$ month (50km)		92.29	1,616,590
cube_key_summary.csv	Cube file statistics		0.00	18

1.3 Species-Level Files (Script 06b)

Table 3: Species-Level Files (from 06b)

type	n_files_by_order	n_files_by_family	total_size_mb_by_order	total_size_mb_by_family
species_summary	2692	654	40.2	14.7
species_cell	2692	654	1582.2	1160.5
species_time	2692	654	1444.7	929.3
species_cell_time	2692	654	7699.1	12483.0

##

Total species-level files: 13384

Total size: 25353.7 MB

1.4 Validation Summary

Table 4: File Validation Summary

check	status	details
Core processed files	PASS	5 / 5 files
Derived core files (06a)	PASS	13 / 13 required
Species by_order files (06b)	PASS	10768 files
Species by_family files (06b)	PASS	2616 files

2. Spatial Coverage Assessment

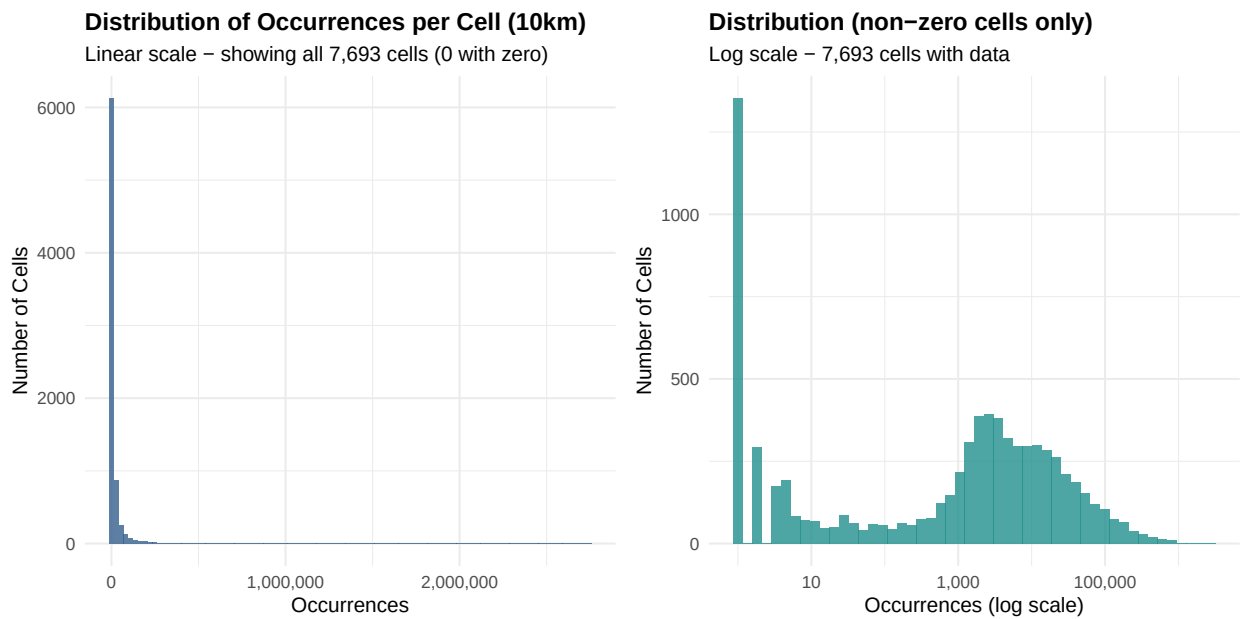
2.1 Coverage Statistics

Table 5: Spatial Coverage Statistics

grid	total_cells	cells_with_data	cells_with_zero_data	best_coverage	total_occurrences	mean	median	sd	co	min	max	q1	q25	q75	q90
10km	7693	7693	0	100	134,239,874	17,450	1820	66,957	1	2,744,153	1	4	9572.03	6394.8	
50km	926	926	0	100	134,239,874	144,967	2	522,114	1	5,856,808	1	1	11690.30	2052.5	

2.2 Occurrence Distribution by Cell

Cell counts: Total: 7693 | With data: 7693 | Zero: 0



2.3 Zero-Coverage Cells (Gap Candidates)

```
## **Zero-coverage cells (10km):** 0 ( 0 %)
```

```
## **Zero-coverage cells (50km):** 0 ( 0 %)
```

2.4 Low-Coverage Cells (Bottom 10%)

```
## **10th percentile threshold:** 1 occurrences
```

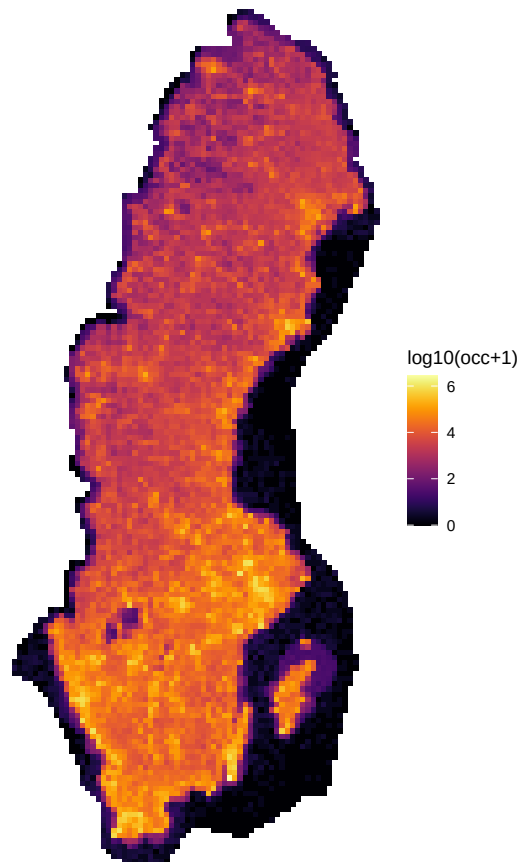
```
## **Cells at or below threshold:** 1353
```

Table 6: Low-Coverage Cells (bottom 10%)

n_cells	min_occ	max_occ	mean_occ
1353	1	1	1

2.5 Spatial Map (if grid available)

Spatial Distribution of Occurrences (10km Grid)
Log scale coloring



2.6 Coverage by Basis of Record

Table 7: Spatial Coverage by Basis of Record (10km)

basisofrecord	n_cells_with_data	total_occurrences	mean_occ
all	7693	134,239,874	17449.6
humanObservation	6303	127,827,874	20280.5
preservedSpecimen	6442	3,784,850	587.5
materialSample	3565	2,543,153	713.4
machineObservation	1451	66,429	45.8
fossilSpecimen	263	7,622	29.0
occurrence	1339	5,574	4.2
observation	38	1,878	49.4
livingSpecimen	469	1,847	3.9
materialCitation	253	647	2.6

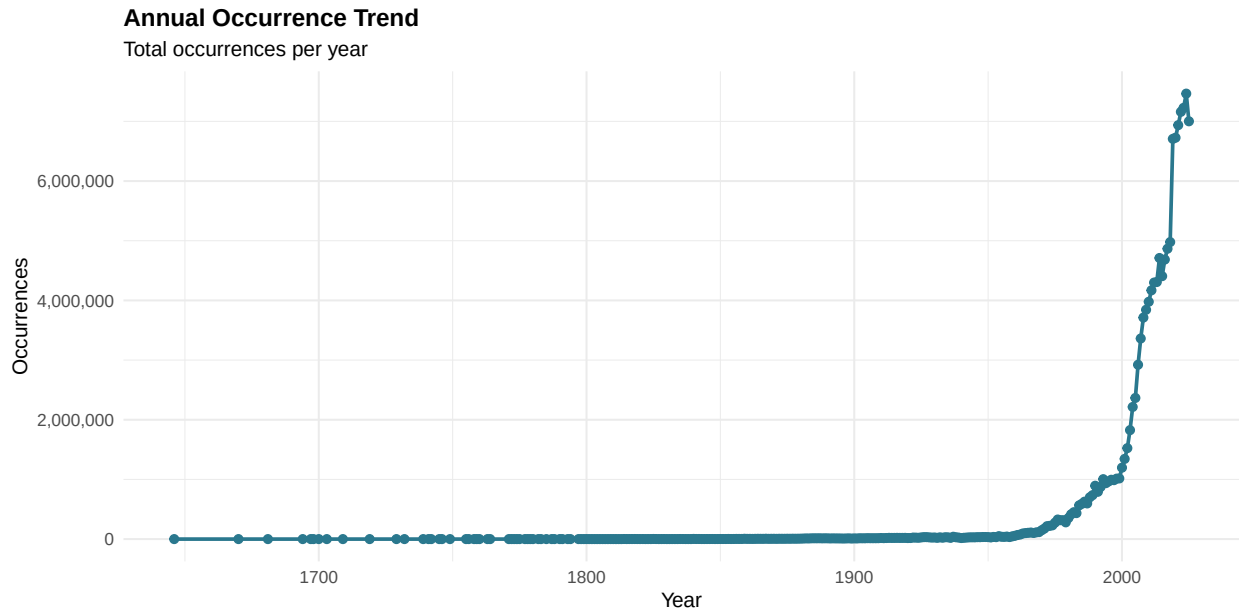
3. Temporal Coverage Assessment

3.1 Temporal Range

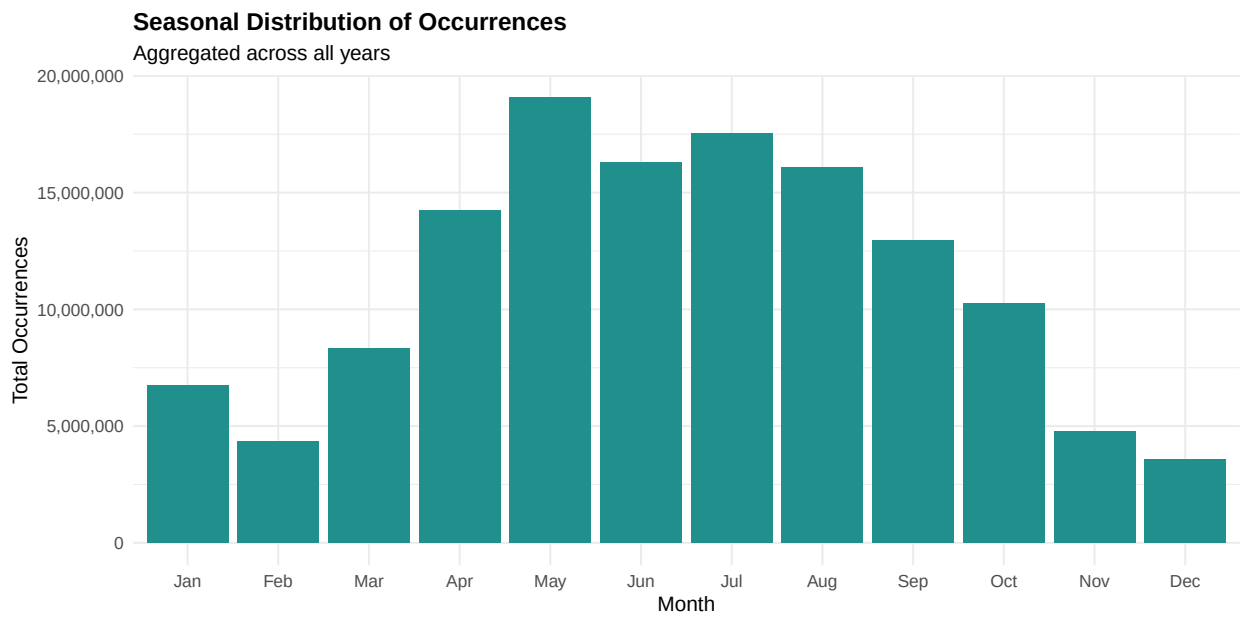
Table 8: Temporal Range Summary

min_year	max_year	n_years	n_months	total_occurrences
1646	2025	272	2668	134239874

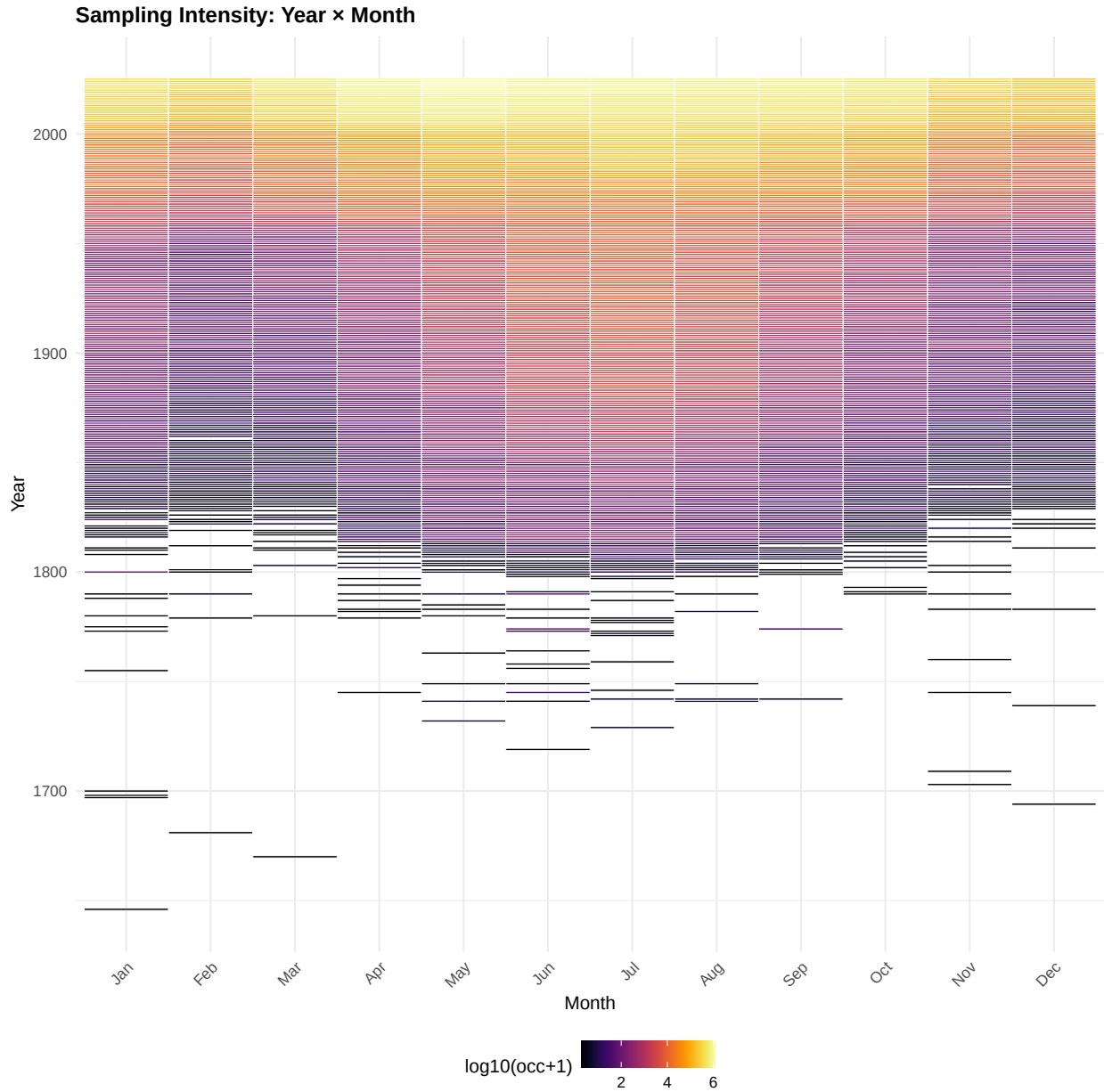
3.2 Annual Trend



3.3 Seasonal Pattern



3.4 Year $\times$ Month Heatmap



3.5 Temporal Gaps

Years with no data: 108

3.6 Recent Data Availability

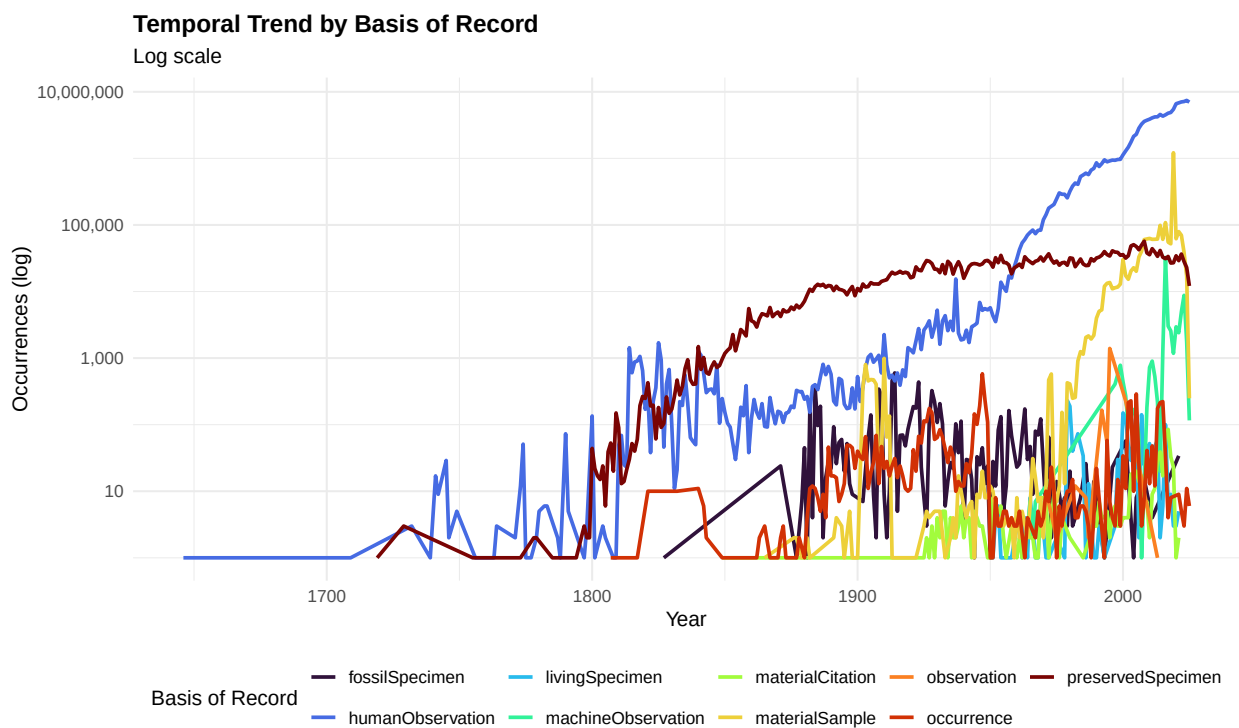
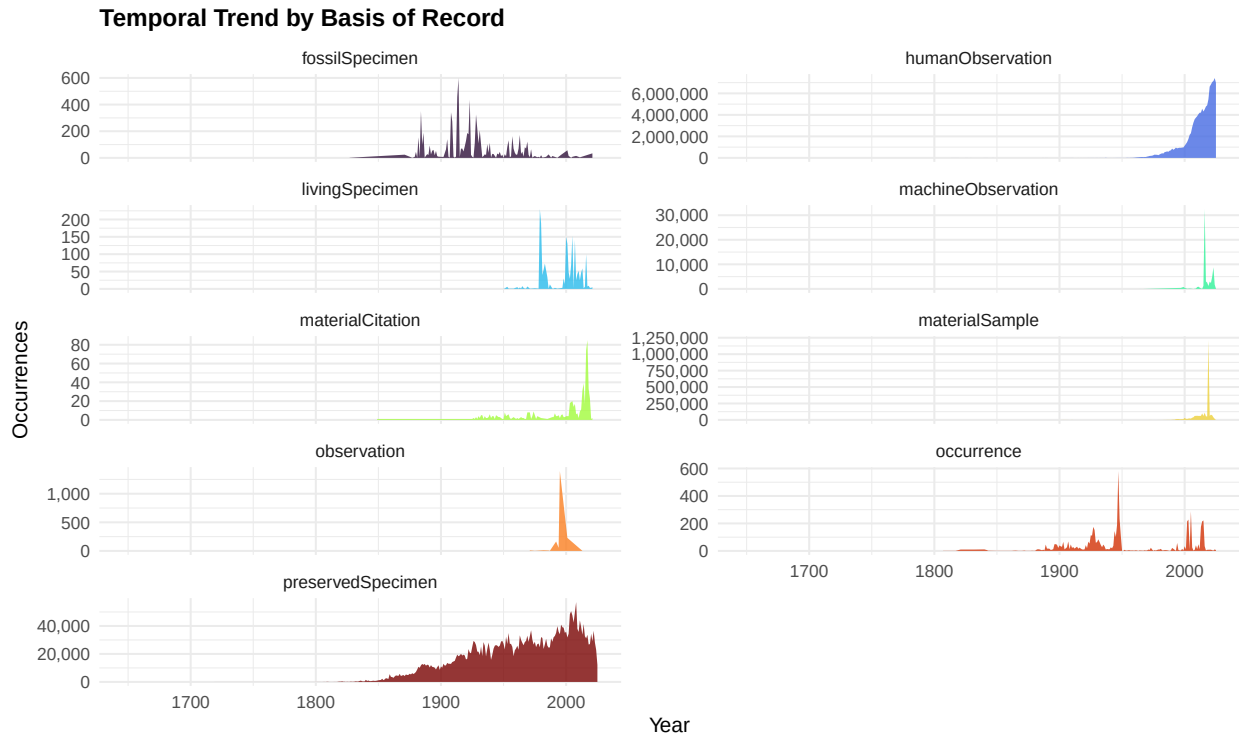
Table 9: Recent Years Data Availability

year	occurrences	n_months	n_cells
2021	6,937,143	12	39750
2022	7,160,302	12	39622
2023	7,226,918	12	39529
2024	7,465,457	12	39850
2025	7,001,677	12	38734

##

Most recent data point: 2025-12-01 with 138,612 occurrences

3.7 Temporal Coverage by Basis of Record



4. Taxonomic Coverage Assessment

4.1 Taxonomic Backbone Summary

Table 10: Taxa by Kingdom

kingdom	n
Animalia	65382
Fungi	18269
Plantae	18006
Chromista	6037
Bacteria	1115
Protozoa	1076
Viruses	107
Archaea	18

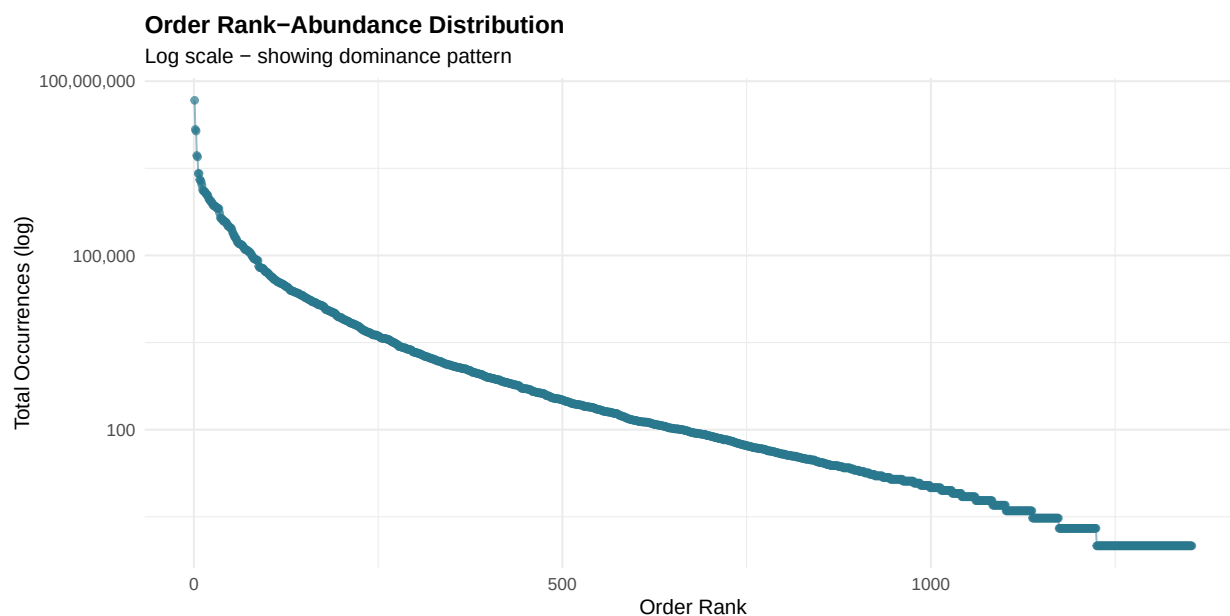
4.2 Order-Level Coverage in GBIF

Number of orders in GBIF data: 1354

Table 11: Top 20 Orders by Occurrence Count

order	total_occurrences	n_months	mean_monthly_occ
Passeriformes	46,793,063	1954	23947
Anseriformes	14,784,576	1663	8890
Charadriiformes	13,866,586	1787	7760
Accipitriformes	5,258,829	1487	3537
Lepidoptera	4,963,034	1229	4038
Piciformes	2,573,334	1174	2192
Poales	2,572,377	1877	1370
Gruiformes	2,010,977	1201	1674
Columbiformes	1,972,035	1089	1811
Asterales	1,826,221	1878	972
Coleoptera	1,610,839	1775	908
Rosales	1,347,404	1680	802
Falconiformes	1,280,829	1192	1075
Podicipediformes	1,273,166	1064	1197
Lamiales	1,254,011	1829	686
Pelecaniformes	1,178,899	1031	1143
Agaricales	1,126,274	1381	816
Diptera	1,095,808	1289	850
Cypriniformes	1,084,404	822	1319
Caryophyllales	958,966	1747	549

4.3 Order Rank-Abundance



4.4 Family-Level Coverage

Number of families in GBIF data: 5023

Table 12: Top 20 Families by Occurrence Count

order	family	total_occurrences	n_months
Anseriformes	Anatidae	14,784,576	1663
Passeriformes	Fringillidae	7,737,540	1391
Charadriiformes	Scolopacidae	5,927,752	1413
Passeriformes	Paridae	5,380,351	1168
Accipitriformes	Accipitridae	4,873,331	1477
Charadriiformes	Laridae	4,855,786	1388
Passeriformes	Corvidae	4,606,036	1330
Passeriformes	Muscicapidae	4,561,368	1196
Passeriformes	Turdidae	3,643,460	1220
Passeriformes	Motacillidae	2,838,502	1174
Passeriformes	Phylloscopidae	2,639,314	851
Piciformes	Picidae	2,573,334	1174
Passeriformes	Sylviidae	2,021,693	911
Columbiformes	Columbidae	1,972,035	1089
Charadriiformes	Charadriidae	1,826,484	1155
Passeriformes	Emberizidae	1,703,507	1135
Asterales	Asteraceae	1,627,032	1867
Passeriformes	Acrocephalidae	1,475,894	610
Passeriformes	Regulidae	1,351,019	894
Poales	Poaceae	1,306,957	1703

4.5 Species-Level Statistics

```
## **Total unique species in GBIF data:** 59,357
## **Total occurrences:** 42,921,177
```

Table 13: Species Count by Order (from species_summary files)

order	n_species	total_occurrences
Diptera	8297	1,095,808
Hymenoptera	6883	804,532
Coleoptera	4812	1,610,839
Agaricales	3245	1,126,274
Hemiptera	1757	321,164
Helotiales	1124	61,021
Unplaced	1083	166,194
Lecanorales	914	713,533
Araneae	797	208,791
Pleosporales	762	13,453
Rosales	682	1,347,404
Ranunculales	583	651,229
Caryophyllales	542	958,966
Pezizales	469	42,320
Russulales	459	381,445

4.6 Threat Status Coverage

```
##
## **Priority threatened taxa (CR/EN/VU/NT):**
## - From backbone: 4547
## - From Red List: 4894
```

5. Bias Assessment

5.1 Basis of Record Bias

```
##
## **Warning:** humanObservation accounts for 95.2% of all occurrences. This may bias gap analysis.
```

5.2 Spatial Sampling Bias

```
## **Debug info:**

## - Rows in cell10_all: 7693

## - Columns: grid, basisofrecord, eeacellcode, occurrences, n_species
```

```

## - Non-zero occurrences: 7693

## **Spatial Gini coefficient:** 0.855
## (0 = perfectly even, 1 = all data in one cell)
##
## **Warning:** High spatial inequality detected. Data is concentrated in few cells.

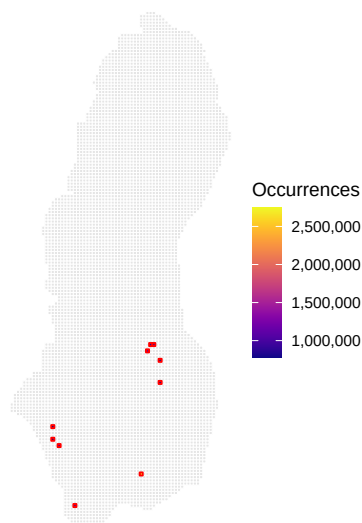
## **Top 10 cells account for:** 8 % of all occurrences

##
## **Top 10 most-sampled cells:**

##      eeacellcode occurrences n_species pct_of_total
##      <char>          <int>    <int>      <num>
## 1: 10kmE471N369      2744153    3547        2.04
## 2: 10kmE473N408      1101005    4611        0.82
## 3: 10kmE475N410      980418     8242        0.73
## 4: 10kmE450N359      946039     4396        0.70
## 5: 10kmE445N378      929408     4671        0.69
## 6: 10kmE474N410      853309     9675        0.64
## 7: 10kmE477N398      801695     2299        0.60
## 8: 10kmE477N405      790770     8194        0.59
## 9: 10kmE443N380      784036     2266        0.58
## 10: 10kmE443N384     783525     8366        0.58

```

Top 10 Most-Sampled Cells
These cells may represent sampling hotspots (cities, research stations)



5.3 Temporal Sampling Bias

```

## **Average year-over-year growth (last 5 years):** 0.9 %

```

5.4 Taxonomic Sampling Bias

Top 5 orders account for: 63.8 % of all occurrences

Top 5 orders: Passeriformes, Anseriformes, Charadriiformes, Accipitriformes, Lepidoptera

6. Gap Analysis Readiness

6.1 Spatial Gap Analysis Readiness

Table 14: Spatial Gap Analysis Readiness

requirement	status	details
Cell summary available		Required for zero-coverage detection
Cell $\times$ time summary available		Required for data recency analysis
Grid geometries available		Required for mapping
Order $\times$ cell available		Required for taxonomic spatial gaps
Zero cells identified	0 cells	Candidates for spatial gaps
Low coverage cells identified	1353 cells	Below 10th percentile

6.2 Temporal Gap Analysis Readiness

Table 15: Temporal Gap Analysis Readiness

requirement	status	details
Time summary available		Required for temporal trends
Cell $\times$ time summary available		Required for cell recency
Order $\times$ time available		Required for taxonomic temporal patterns
Family $\times$ time available		Required for family-level trends
Year range adequate		272 years of data
Recent data available		Most recent: 2025

6.3 Taxonomic Gap Analysis Readiness

Table 16: Taxonomic Gap Analysis Readiness

requirement	status	details
Taxa backbone available		National taxonomic backbone reference
Threat status (backbone)		60527 taxa
Threat status (Red List)		6070 taxa
Species summaries available		1346 order files
Bias correction columns		familycount, ordercount, classcount in species tables

6.4 Overall Readiness Score

```
## ## Overall Gap Analysis Readiness
```

```
## **Checks passed:** 17 / 17 ( 100 %)
```

```
##  **READY** for comprehensive gap analysis
```

7. Recommendations

```
## Based on this sanity check, consider the following:
```

```
##
```

```
## - Consider spatial bias correction due to high sampling inequality
```

```
## - 108 years have no data - may affect trend analysis
```

```
## - ~134,791 Backbone taxa may be missing from GBIF - run taxonomy matching
```

8. Data Quality Summary

Table 17: Data Quality Summary

dimension	metric	value	status
Files	Core files present	5/5	
Files	Derived files present	13/13	
Spatial	Cell coverage	100%	
Spatial	Zero-coverage cells	0	
Temporal	Year range	1646-2025	
Temporal	Data recency	2025	
Taxonomic	Orders covered	1,354	
Taxonomic	Families covered	5,023	
Bias	Spatial Gini	0.85	
Bias	Top 5 orders share	63.8%	

Session Info

```
## R version 4.5.2 (2025-10-31)
```

```
## Platform: aarch64-apple-darwin20
```

```
## Running under: macOS Tahoe 26.2
```

```
##
```

```
## Matrix products: default
```

```
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/
```

```

## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Stockholm
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices datasets  utils      methods    base
##
## other attached packages:
## [1] targets_1.11.4      httr2_1.2.2          leaflet_2.2.3        plotly_4.12.0        shinyWidgets_0.9.0
## [7] patchwork_1.3.2      DT_0.34.0            tibble_3.3.1         forcats_1.0.1        knitr_1.50
## [13] viridisLite_0.4.2    ggplot2_4.0.1        scales_1.4.0         lubridate_1.9.4      fst_0.9.8
## [19] tidyr_1.3.2          data.table_1.18.0    stringr_1.6.0        readr_2.1.6          dplyr_1.1.4
## [25] cli_3.6.5            glue_1.8.0           purrr_1.2.1          here_1.0.2
##
## loaded via a namespace (and not attached):
## [1] DBI_1.2.3              gridExtra_2.3          rlang_1.1.7           magrittr_2.0.4
## [6] e1071_1.7-17           compiler_4.5.2         systemfonts_1.3.1     callr_3.7.6
## [11] pkgconfig_2.0.3        crayon_1.5.3           fastmap_1.2.0         backports_1.5.0
## [16] labeling_0.4.3         utf8_1.2.6            promises_1.5.0        rmarkdown_2.30
## [21] ps_1.9.1               tinytex_0.58           bit_4.6.0             xfun_0.56
## [26] jsonlite_2.0.0         later_1.4.5            tarchetypes_0.13.2    terra_1.8-93
## [31] prettyunits_1.2.0      R6_2.6.1              bslib_0.9.0           stringi_1.8.7
## [36] jquerylib_0.1.4        Rcpp_1.1.1            leaflet.providers_2.0.0 httpuv_1.6.16
## [41] timechange_0.3.0       tidyselect_1.2.1       rstudioapi_0.18.0     yaml_2.3.10
## [46] curl_7.0.0             processx_3.8.6         withr_3.0.2           S7_0.2.1
## [51] units_1.0-0            proxy_0.4-29           xml2_1.5.2            pillar_1.11.1
## [56] KernSmooth_2.23-26     renv_1.1.6            generics_0.1.4        vroom_1.6.7
## [61] hms_1.1.4             xtable_1.8-4          base64url_1.4         class_7.3-23
## [66] tools_4.5.2           fs_1.6.6              grid_4.5.2            crosstalk_1.2.2
## [71] kableExtra_1.4.0       textshaping_1.0.4     svglite_2.2.2         gt_1.3.0
## [76] sass_0.4.10            digest_0.6.37          classInt_0.4-11       htmlwidgets_1.6.4
## [81] memoise_2.0.1          htmltools_0.5.8.1     lifecycle_1.0.4       httr_1.4.7
## [86] mime_0.13             bit64_4.6.0-1

```